Physics - Undergraduate Level Assessment 19 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons

   (a) an average of 10 times, with an rms deviation of about 4

   (b) an average of 10 times, with an rms deviation of about 3

   (c) an average of 10 times, with an rms deviation of about 1

   (d) an average of 10 times, with an rms deviation of about 0.1

2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?

   (a) 2

   (b) 250

   (c) 5,000

   (d) 10,000

3. The emission spectrum of the doubly ionized lithium atom Li++ (Z = 3, A = 7) is identical to that of a hydrogen atom in which all the wavelengths are

   (a) decreased by a factor of 9

   (b) decreased by a factor of 49

   (c) decreased by a factor of 81

   (d) increased by a factor of 9

4. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately

   (a) 0.1 GeV/c^2

   (b) 0.2 GeV/c^2

   (c) 0.5 GeV/c^2

   (d) 1.0 GeV/c^2

5. The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?

   (a) Specific heat

   (b) Thermal conductivity

   (c) Electrical resistivity

   (d) Hall coefficient

True / False (10 marks)

Answer True or False

6. True or False: In a constant pressure process, the increase in the internal energy of an ideal gas is equal to the heat added to the gas.
7. True or False: In the diamond structure of elemental carbon, the nearest neighbors of each C atom are located at the corners of a tetrahedron.
8. True or False: The total spin quantum number of the electrons in the ground state of neutral nitrogen is 1.
9. True or False: Fermions have antisymmetric wave functions and obey the Pauli exclusion principle, preventing them from occupying the same quantum state.
10. True or False: An atom with filled n = 1 and n = 2 levels contains only 6 electrons based on the maximum capacity of these energy levels.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A particle decays in its rest frame in 2.0 ms. Analyze how its decay time changes when moving at 0.60 c in the lab frame and calculate the distance it travels before decaying.
12. Discuss the relationship between the absolute temperature of a blackbody and the energy it radiates per second per unit area, particularly when the temperature is increased by a factor of 3.

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